

Umesha Rathnayake

SOFTWARE ENGINEER INTERN

+94 78 524 6978 | umesharandini24@gmail.com | [GitHub](#) | [LinkedIn](#)

About Me

I am an enthusiastic Information Technology undergraduate with a strong interest in software engineering and technology. I enjoy building practical projects, learning new technologies, and solving problems through programming. I have experience working on web applications, software projects, and hardware-based projects as part of university work. I am a quick learner who enjoys collaborating with others as a team. I am currently looking for a software engineering internship where I can improve my skills, gain real-world experience, and contribute to meaningful projects.

Education

University of Moratuwa

March 2024–Present

Bachelor of Science Honours in Information Technology (GPA: 3.10 / 4.00)

- **Relevant Coursework:** Data Structures and Algorithms I & II (C, Java), Object-Oriented Programming (Java), Software Engineering, Operating Systems, and Computer Architecture

Bandaranayake Central College, Veyangoda

2022

Physical Science Stream

- **G.C.E. Advanced Level 2022 - Physical Science Stream:**
 - * **Z-score:** 1.5953
 - * **Results:** ABB(A-Combined Mathematics)

Projects

User Management System | TypeScript, Next.js, Java 21, Spring Boot, PostgreSQL, AWS

- **Company:** Informatics International Ltd.
- Developed a centralized web-based User Management System with authentication, role-based access control (RBAC), and location-based access management.
- Built the frontend using Next.js, TypeScript, Zustand, Tailwind CSS, shadcn/ui, React Hook Form, and Zod.
- Developed RESTful backend services using Java 21 and Spring Boot with JWT-based authentication and OAuth 2.0 support.
- Implemented PostgreSQL for relational data management and AWS for secure user profile file storage.
- Collaborated with the development team using Git and contributed to frontend development, database design, authentication, and RBAC implementation.

SonicGlowWeb: 8×8×8 LED Cube Control System | C++, JavaScript, React, Node.js, Hardware Integration

- Designed and implemented a hardware-software bridge for a 512-LED cube using NodeMCU (ESP8266) and Arduino microcontrollers, addressing GPIO limitations through 74HC595 shift-register multiplexing.
- Designed PCB routing and circuit architecture for 512 LEDs using layer- and column-based control logic, enabling efficient power distribution and reliable signal transmission.
- Developed a React-based web frontend with interactive 3D animation and pattern-control interfaces for real-time LED cube management.
- Built a Node.js backend to establish communication between the web application and microcontrollers, enabling real-time command transmission and device control.
- Collaborated with team members using Git and maintained modular code organization for scalable hardware-software integration.

Personal Portfolio Website | React, Vite, JavaScript, CSS, Framer Motion | [Umesha Rathnayake](#)

- Designed and developed a modern, responsive personal portfolio website to showcase projects, skills, certifications, and extracurricular activities.
- Built using React with Vite for fast development and optimized production builds, featuring component-based architecture with CSS Modules for scoped styling.
- Implemented smooth scroll-reveal animations using Framer Motion, interactive UI elements, and a glassmorphism-inspired dark theme for a premium user experience.
- Deployed on Vercel with CI/CD integration via GitHub for automatic redeployment on every push.

Technical Skills

Programming Languages: Java, C, Python, JavaScript, TypeScript

Frontend Development: React.js, Next.js, HTML5, CSS3

Backend Development: Spring Boot, Node.js, Express.js, RESTful APIs

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Object-Oriented Design, Fundamentals of Software Engineering

Networking & Systems: Computer Networks, TCP/IP, Socket Programming, Operating Systems, Computer Architecture

Databases: PostgreSQL, MySQL, SQL

Security & Authentication: JWT, Role-Based Access Control (RBAC), Authentication & Authorization

Tools & Platforms: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Maven

Machine Learning: Supervised Learning, Regression, Classification

Experience

Smart Pick Pack (Pvt) Ltd.

April 2023 – February 2024

Junior Customer Relationship Executive

Sri Lanka

- * Worked as a Junior Customer Relationship Executive for 11 months, supporting customer communication and relationship management.
- * Handled customer inquiries, maintained professional communication, and supported day-to-day customer service activities.
- * Developed strong communication, teamwork, problem-solving, and customer service skills in a professional working environment.

Extracurricular Activities

Women's Elle Team

2025–Present

Player

University of Moratuwa

IEEE Women in Engineering (WIE)

2025–Present

Committee Member – Marketing Committee & Other University Coordinator

University of Moratuwa

IEEE Robotics and Automation Society (RAS)

2025

Volunteer

University of Moratuwa

IESL Laboratory (Software Branch)

2025–Present

Member

Faculty of Information Technology, University of Moratuwa

Classical Music Society

2025–Present

Member

University of Moratuwa

Awards & Certifications

University Colours Award – 2025

2025

Women's Elle

University of Moratuwa

SESERVE Mora 2024 – Table Tennis

2024

1st Runner-Up

University of Moratuwa

Certificate of Appreciation – FITExpo 2025

2025

Microcontroller-Based Project

Faculty of Information Technology, University of Moratuwa

Certificate of Appreciation – Beyond The Pages 1.0

2025

Volunteer

IEEE Robotics and Automation Society, University of Moratuwa

Referees

Hashitha Keerthi

Software Engineer

- * Email: hashitha.keerthi@ifs.com
- * Mobile: +94 76 356 5869

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